

What Is Automation?

- **Automation** means making a process **run automatically without human effort**, using tools, scripts, or machines.
- In software, **automation** is about **using technology to perform repetitive tasks** - such as testing, deployment, or data processing - **quickly, accurately, and consistently**.

Simple Definition

Automation is the process of using software tools or scripts to perform tasks that would otherwise require manual human effort.

Example

- **Manual:** You click “Login” and type credentials every time.
- **Automation:** You write a script that **automatically opens the browser, fills in credentials, clicks login, and checks if login succeeded**.

Types of Automation in Software Development

Type	Purpose	Tools / Technologies
Test Automation	Automate functional, regression, or API tests	Selenium, Playwright, Cypress, Appium
Build Automation	Compile & build code automatically	Maven, Gradle, Webpack
Deployment Automation	Deploy apps to servers or cloud automatically	Jenkins, Docker, Kubernetes
CI/CD Automation	Continuous Integration & Delivery pipelines	GitHub Actions, GitLab CI, Azure DevOps
Process Automation	Automate business workflows	UiPath, Power Automate, Blue Prism
Performance Automation	Automate load/stress tests	JMeter, LoadRunner

Automation in Testing

Automation Testing (most common type) means:

- Writing **test scripts** to check if your app behaves correctly.
- Running tests **automatically** after each build or code change.
- Saving time by avoiding manual repetition.

Example using Playwright (TypeScript):

```
1 import { test, expect } from '@playwright/test';
2 test('automated login test', async ({ page }) => {
3   await page.goto('https://example.com/login');
4   await page.fill('#username', 'admin');
5   await page.fill('#password', 'admin123');
6   await page.click('#loginButton');
7   await expect(page).toHaveTitle(/Dashboard/);
8 });
```

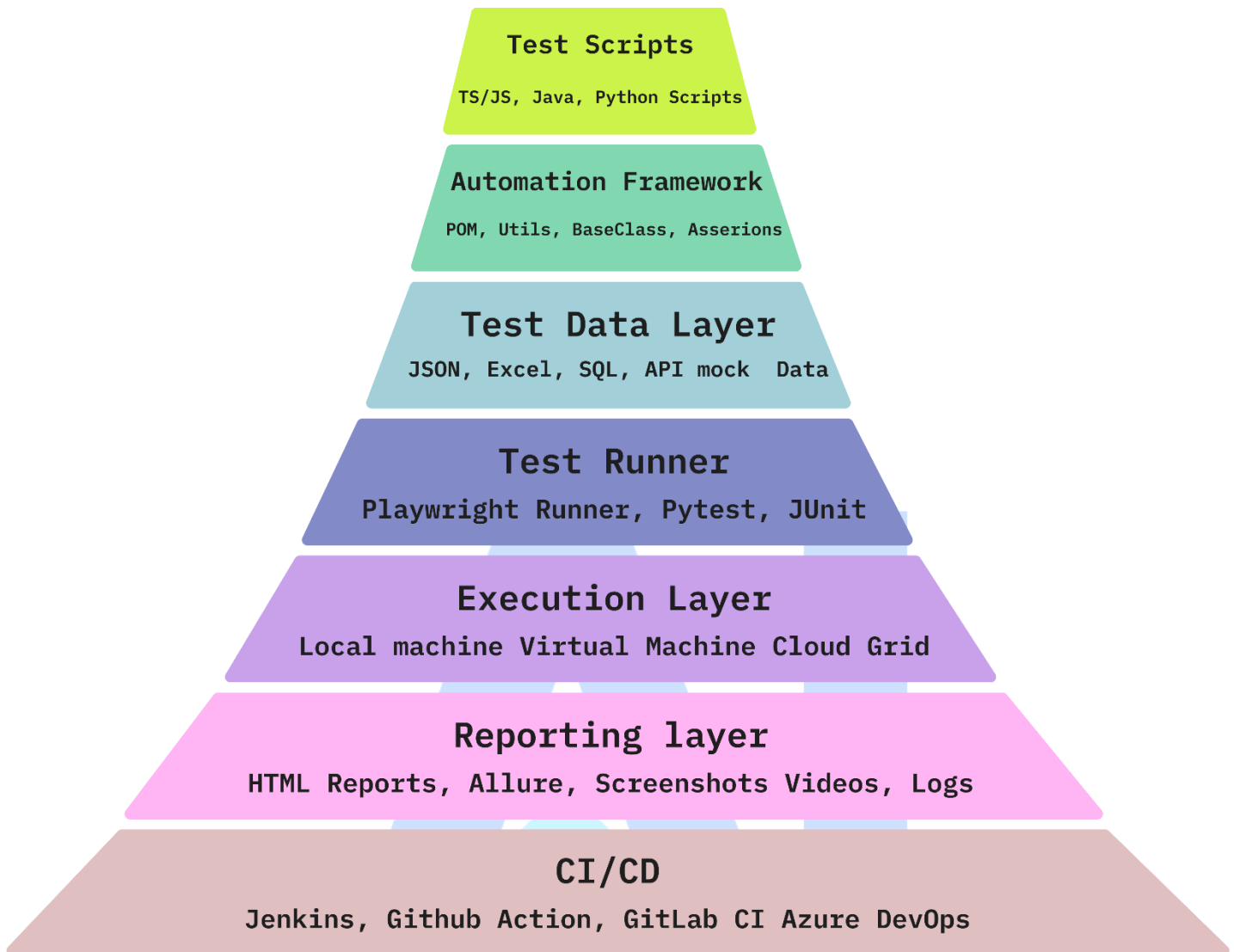
Why Automation Is Important

BENEFIT	EXPLANATION
SPEED	Executes hundreds of test cases in minutes
ACCURACY	Eliminates human errors
EFFICIENCY	Frees testers from repetitive work
REUSABILITY	Scripts can run anytime on any environment
SCALABILITY	Run parallel tests on multiple browsers/devices
CI/CD READY	Integrates into pipelines for continuous testing

Automation Architecture (in Testing)

- 1. **Test Scripts:** Your automated test cases (TypeScript, Python, Java).
- 2. **Framework:** Provides structure (Playwright, Selenium, Cypress).
- 3. **Test Data:** Input values from JSON, Excel, DB.
- 4. **Test Runner:** Executes tests (Playwright Test Runner, PyTest, JUnit).
- 5. **Reports:** Auto-generated summaries (HTML, JSON).
- 6. **CI/CD Integration:** Automatically triggered on every code push.

Automation architecture diagram



Automation Lifecycle

1. **Select tools** – e.g., Playwright for web testing.
2. **Plan automation scope** – what to automate (login, API, regression).
3. **Create framework** – POM, reusable libraries.
4. **Write scripts** – in TypeScript/Python.
5. **Execute tests** – via CLI or CI/CD.
6. **Review reports** – HTML, Allure, etc.
7. **Maintain tests** – update when app changes.

Example: End-to-End Automation

You're testing an e-commerce website:

1. Login → Add item to cart → Checkout → Logout
2. All these steps are automated using Playwright scripts.
3. Every time the developer updates the code, Jenkins runs your automation suite.
4. If a bug appears, the test fails, and a report with screenshots is emailed automatically.

Note: This is complete automation.

Automation in One Line

Automation is the use of technology to execute tasks automatically - improving speed, accuracy, and efficiency while reducing manual effort



